Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
6	(a)			**Indicative content:** In the fission reaction 1 neutron is absorbed by each uranium nucleus causing fission which gives rise to 3 further neutrons. These can then cause further fissions and even more neutrons so the reaction is uncontrolled. The moderator slows down fast-moving neutrons. This is so that they can be absorbed by the uranium nuclei and allow fission to occur. Control rods absorb neutrons so that on average each fission reaction leads to one further fission. They can be raised and / or lowered to control the rate of the reaction. **5 – 6 marks** Comprehensive description of all 3 areas. *There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.* **3 – 4 marks** Comprehensive description of 2 of the 3 areas **or** limited description of all 3. *There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.* **1–2 marks** Limited description of 1 or 2 of the 3 areas. *There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.* **0 marks** *No attempt made or no response worthy of credit.*	6			6		

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Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)		${}^{2}_{1}\mathrm{H}(1) + {}^{3}_{1}\mathrm{H}(1) \rightarrow {}^{4}_{2}\mathrm{He} + {}^{1}_{0}\mathrm{n}(1)$ for the RHS		3		3		
		(ii)		Waste products not radioactive **or** more energy produced [per reaction] **or** fuel readily available **or** no chance of an uncontrolled reaction		1		1		
				Question 6 total	**6**	**4**	**0**	**10**	**0**	**0**

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